

Problem

A commercially available three-phase inductive motor operates at a full load of 120 hp (1 hp = 746 W) at 95 percent efficiency at a lagging power factor of 0.707. The motor is connected in parallel to a 80-kW balanced three-phase heater at unity power factor. If the magnitude of the line voltage is 480 V rms, calculate the line current.

Step-by-step solution

Step 1 of 3

The power output of the motor is,

$$\begin{aligned}P_{\text{out}} &= 120 \text{ hp} \\&= (120)(746 \text{ W}) \\&= 89520 \text{ W}\end{aligned}$$

Calculate the input power.

$$\begin{aligned}P_{\text{in}} &= \frac{P_{\text{out}}}{\eta} \\&= \frac{89520}{0.95} \\&= 94231.57 \text{ W}\end{aligned}$$

The power factor is,

$$\begin{aligned}\cos \theta_1 &= 0.707 \\ \theta_1 &= \cos^{-1}(0.707) \\ &= 45^\circ\end{aligned}$$

Calculate the reactive power.

$$\begin{aligned}Q_1 &= P_1 \tan \theta_1 \\&= (94231.57) \tan 45^\circ \\&= 94231.57 \text{ VAR}\end{aligned}$$

Step 2 of 3

Therefore, the complex power provided by the motor is,

$$\begin{aligned} S_1 &= P_1 + jQ_1 \\ &= 94.232 + j94.232 \text{ kVA} \end{aligned}$$

Calculate the complex power absorbed by the three-phase heater.

$$\begin{aligned} S_2 &= P_2 + jQ_2 \\ S_2 &= \frac{P_2}{\cos \theta_2} \\ &= \frac{80 \text{ kW}}{1} \\ &= 80 \text{ kVA} \end{aligned}$$

The reactive power is,

$$\begin{aligned} Q_2 &= S_2 \sin \theta_2 \\ &= 80(0) \\ &= 0 \text{ kVA} \end{aligned}$$

Therefore, the complex power absorbed by the three-phase heater is,

$$\begin{aligned} S_2 &= P_2 + jQ_2 \\ &= 80 \text{ kVA} \end{aligned}$$

Step 3 of 3

Determine the total complex power.

$$\begin{aligned} S &= S_1 + S_2 \\ &= 94.232 + j94.232 + 80 \text{ kVA} \\ &= 174.232 + j94.232 \text{ kVA} \\ &= 198.082 \angle 28.41^\circ \text{ kVA} \end{aligned}$$

Calculate the line current.

$$\begin{aligned} |S| &= \sqrt{3} V_L I_L \\ I_L &= \frac{|S|}{\sqrt{3} V_L} \end{aligned}$$

Substitute 198.082 kVA for $|S|$ and 480 V rms for V_L .

$$\begin{aligned} I_L &= \frac{|S|}{\sqrt{3} V_L} \\ &= \frac{198.082 \text{ kVA}}{(\sqrt{3})(480 \text{ V rms})} \\ &= \frac{198.082 \text{ kVA}}{831.384 \text{ Vrms}} \\ &= 238.255 \text{ A} \end{aligned}$$

Therefore, the line current, I_L is 238.255 A.